Band-1 Hadamard Order 2804 — Miyamoto (H_700 Paley-II) + Paley-I Cross-Check; Gap 2052 Still OPEN

H_700/C_702/H_2804 PASS (Miyamoto + Paley-I); 2804 in SPK via Paley I q=2803; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_2708_Miyamoto`; `2026-09-15_Band1_Hadamard_2692_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_2804_miyamoto/`.
Cutoff. 2026-09-15 (~02:39Z).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 700 and 2804 only** (verified max $|HH^\top - nI| = 0$), via **two independent paths** for order 2804 (Miyamoto primary; Paley type I cross-check). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 2804 **enters SPK** via Paley I ($q = 2803$); it is **not** a Miyamoto-only bonus (contrast prior 2692/2564/2276/2228/2164). Honest SPK audit: Paley-II **FAIL** ($q = 1401$ not prime); Kronecker **FAIL** ($2804 = 4 \cdot 701$, 701 not Hadamard-admissible); Paley-I **PASS**. `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 2708 on box. This increment constructs the **next preferred** Miyamoto-eligible target after $q = 677$: **order 2804** ($2804 = 4 \cdot 701$, 701 prime $\equiv 1 \pmod 4$). Seed H_{700} via Paley type II ($q = 349$ prime $\equiv 1 \pmod 4$; order $2(q+1) = 700$). Symmetric Paley conference C_{702} over prime 701. $H_{2804} = \text{Miyamoto}(H_{700}, C_{702})$. Independently, H_{2804} also verifies via **Paley type I** with $q = 2803$ (explains SPK membership). Unit tests **26/26 PASS**. SPK next gap remains **2052** (OPEN). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2804; Miyamoto; Paley type II; Paley type I; Paley conference; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{700} constructed and verified (Paley type II, $q = 349$). 2. C_{702} constructed and verified (Paley conference, $q = 701$): $CC^\top = 701I$, zero diagonal. 3. H_{2804} constructed and verified (Miyamoto). **Also** constructed and verified via Paley type I ($q = 2803$). **Not** the Hadamard conjecture. 4. SPK: $2804 \in$ SPK (via Paley I). Paley-II and Kronecker paths for 2804 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$2804 = 4 \cdot 701$
$q = 701$	prime, $701 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{700}
Needed conference	order $702 = q + 1$
H_{700} path (used)	Paley type II, $q = 349$ (prime $\equiv 1 \pmod 4$); $2(q+1) = 700$
Conference path	Symmetric Paley over prime 701
SPK path for 2804	Paley type I, $q = 2803$ prime $\equiv 3 \pmod 4$; order $q + 1 = 2804$
Paley-II for 2804	**REJECTED**: $q' = 1401$ not prime
Kronecker for 2804	**REJECTED**: only 4×701; $701 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{700}	Paley type II, $q = 349$	0	0.2186	**PASS**
C_{702}	Paley conference, $q = 701$	0	0.4431	**PASS**
H_{2804}	Miyamoto(H_{700}, C_{702})	0	9.5053	**PASS**
H_{2804} P-I	Paley type I, $q = 2803$	0	13.7074	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh + Paley-I cross-check) ≈ 24.9234 s. Compressed ± 1 matrices saved as `H700_pm1.npz`, `C702_pm1.npz`, `H2804_pm1.npz`, `H2804_paleyI_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships $\cup$ verified $H_{700} \cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 2804 enters SPK **because** Paley I applies ($q = 2803$).

Quantity	Value
Known count	882
2804 in SPK	**True** (Paley I $q = 2803$)
700 in SPK	True (Paley II $q = 349$)
2708 in SPK	True (prior Paley I)
2692 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284...

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (26/26)

Test	Result
`H4_sylvester`	PASS
`H20_paley_I_control`	PASS
`paley_II_q5_H12`	PASS

Test	Result
paley_II_q13_H28	PASS
miyamoto_H20_from_H4_conf6	PASS
paley_I_q11_H12	PASS
q349_is_prime	PASS
q349_equiv_1_mod_4	PASS
H700_paley_II_order_formula	PASS
q701_is_prime	PASS
q701_equiv_1_mod_4	PASS
n2804_equals_4_times_701	PASS
H700_order_formula	PASS
C702_order_formula	PASS
paley_I_H2804_path_eligible	PASS
paley_II_H2804_path_rejected	PASS
kronecker_H2804_path_rejected	PASS
q2803_is_prime_equiv_3_mod_4	PASS
paley_I_order_2804_formula	PASS
reject_all_ones_H8	PASS
conference_6_paley_control	PASS
miyamoto_eligible_2804_arith	PASS
conference_702_shape_expected	PASS
gap_2052_not_filled_by_this_ship	PASS
2804_in_spk_via_paley_I_arith	PASS
paley_II_seed_pattern_control	PASS

7. Next optional targets

Miyamoto-eligible primes $q > 701$, $q \equiv 1 \pmod 4$ (sample):

q	$n = 4q$	Need H_{q-1}	On-box path notes
709	2836	708	PaleyII(q=353)
733	2932	732	—
757	3028	756	—
761	3044	760	Double(PaleyI half=380), Kronecker(380x2)
769	3076	768	Double(PaleyI half=384), Kronecker(4x192)
773	3092	772	—
797	3188	796	PaleyII(q=397)
809	3236	808	Kronecker notes

Preferred next. Method-attack remaining SPK gap **2052** (GS/SDS for order ~513 if literature

seeds appear), **or** construct Miyamoto $q = 709 \to n = 2836$ (seed Paley-II $q = 353$), **or** Track B privacy / Object 1 meter await. GATE-04 remains LOCKED.

8. Non-claims (explicit)

- Not the full Hadamard conjecture.
- Not Clay-closed / finished-ToE / RH / P≠NP / YM / Frankl / NS-solved / AC-solved.
- Gap **2052** remains **OPEN**.
- Order 2804 is **in SPK** (Paley I); not a Miyamoto-only bonus.
- GATE-04 collection **LOCKED**; no invented J.
- `clay_closed = false`.

9. Artifacts

Lab: `/workspace/experiments/band1/hadamard_2804_miyamoto/`
Verifier: `verify_hadamard_2804.py`
Results: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
Matrices: `H700_pm1.npz`, `C702_pm1.npz`, `H2804_pm1.npz`, `H2804_paleyI_pm1.npz`
Shelf: `/workspace/DOWNLOAD_PDFs/2026-09-15_Band1_Hadamard_2804_Miyamoto.pdf`